

Formal Legal Abduction: Extracting Implicit Assumptions in Supreme Court FIRAC Appeals

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Abstract

Legal reasoning essentially relies on unstated background rules, commercial standards and intermediate factual links that can leave a logical gap between explicit facts, governing statutes, and final legal judgements. Although Large Language Models (LLMs) are promising in legal text processing, existing reasoning approaches suffer from an "over-conservatism bias", due to which models routinely accept authoritative judicial phrasing as theoretically valid without detecting implicit assumptions. To address this challenge, this dissertation presents an automated, high-sensitivity abductive reasoning pipeline model designed to extract and formally ground implicit legal premises from judicial appeal decisions. Working on the FIRAC-Appeals dataset of Indian Supreme Court judgements, the proposed framework evaluates sequential application claims against structured FIRAC facts and rules.

Through iterative prompt engineering and schema hardening, the pipeline resolves baseline model over-conservatism when evaluated across the full 250-case evaluation cohort of 3,534 reasoning steps in 250 appeal cases, achieving an 89.64% SUPPORTED classification rate and flagging 252 steps (7.13%) as UNSUPPORTED. 114 steps (3.23%) required abductive premise extraction, of which 92.1% were successfully constrained to the closed 7-predicate ontology defined in Chapter 3. To overcome the "abduction-space problem" – where unconstrained LLMs generate arbitrary free-text assumptions, the extracted premises are constrained to canonical predicate triples ([Subject] -> [Predicate] -> [Object]) aligned with the Staram Indian Legal Ontology. Finally, a critical audit of the dataset reveals four systematic structural flaws (such as rule circularity and fact summarization artifacts) establishing a foundation for a re-annotated benchmark dataset ("FIRAC-Abduce"). This work demonstrates the feasibility of combining large language models with structured legal ontologies to enable traceable, neuro-symbolic legal abduction.

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Furthermore, I would like to thank my peers and fellow researchers for their constructive discussions and encouragement throughout this master's programme.

Finally, I am deeply thankful to my family for their constant encouragement and support.

Generative AI Declaration

In accordance with University of Glasgow guidelines, I declare the following use of Artificial Intelligence (AI) tools in the preparation of this dissertation:

- Drafting and Proofreading: AI tools were utilized for grammatical proofreading, stylistic refinement, and formatting assistance during the drafting of this dissertation.
- Code Assist and Debugging: Generative AI tools were used to assist in structuring python scripts, formulating API request logic, and designing JSON extraction schemas for the data pipeline. All code was verified, tested, and modified by the author.
- Experimental Execution: As detailed in Chapters 3 and 4, Large Language Models (qwen-2.5-coder-7b-instruct and nli-deberta-v3-base) served as the primary subjects of empirical testing and formed the core experimental methodology for this research.

I take full personal responsibility for the accuracy, originality, and validity of all content, code, and conclusions presented in this work.

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Chapter 1: Introduction

Legal reasoning in court decisions is rarely built as clear, explicit mathematical deductions. In practice, judges often leave out intermediate logical steps, background statutory rules, and commercial trade standards when delivering judgements. Rather than articulating every logical link from scratch, judicial arguments are structured as enthymemes (Walton, 2001, 2008) – syllogistic deductions that rely on implicit, unstated premises assumed to be understood as common legal or factual knowledge.

While modern Large Language Models (LLMs) demonstrate remarkable capabilities in natural language understanding, evaluating their utility as automated legal verifiers reveals a widespread vulnerability: over-conservatism bias (Guha et al., 2023; Katz et al., 2024). This is the systematic tendency of neural language models to evaluate authoritative, formal, or high-register legal prose as deductively complete, granting the author the "benefit of the doubt" and failing to detect missing intermediate premises required to make the judicial conclusion formally valid. In high-stakes computational law systems, this failure leaves critical logical discontinuities and unstated statutory assumptions unnoticed.

To address this challenge, this dissertation develops an automated, ontology-grounded abductive legal reasoning pipeline. By evaluating structured FIRAC (Facts, Issues, Rules, Application, Conclusion) representations of Indian Supreme Court appeal decisions, the system moves beyond binary classification by extracting, categorizing and formalizing the exact missing premises required to bridge judicial arguments.

The primary research objectives of this dissertation are:

- Abductive Reasoning Architecture: To develop a formal verification methodology using large language models capable of auditing multiple-step legal application claims against explicit rules and facts to reconstruct unstated premises.
- Ontological Grounding: To constrain the unconstrained “abduction-space problem” by enforcing mutually exclusive taxonomy categories and structuring extracted assumptions into canonical predicate triples ([Subject] -> [Predicate] -> [Object]) aligned with the Staram Indian Legal Ontology.
- Empirical Benchmarking: To benchmark verification performance across a 250-case, 3,534-step evaluation cohort drawn from the FIRAC-Appeals dataset[1], measuring classification outcomes, identifying the extent to which dataset-level artifacts constrain genuine implicit-premise extraction, and evaluating schema compliance against the closed 7-predicate ontology.
- Benchmark Auditing: To conduct a systematic audit of the FIRAC-Appeals dataset benchmark, identifying structural artifacts (such as rule circularity and fact summarization gaps) to inform future legal AI benchmark construction (“FIRAC-Abduce”).

[1] Available at: https://huggingface.co/datasets/mborcin/firac-appeals

1.1 Task Formulation

We formalize the task of legal premise extraction as an abductive reasoning problem over structured FIRAC components:

Given:

- A set of explicit facts $F = \{f_1, f_2, \ldots, f_m\}$.
- A set of explicit rules $R = \{r_1, r_2, \ldots, r_n\}$.
- A sequential judicial application claim C_i,

The verifier evaluates the formal logical status, S: {SUPPORTED, NEEDS_ASSUMPTION, UNSUPPORTED} according to the following criteria:

1. SUPPORTED: The claim C_i is strictly entailed by explicit context alone:
$$(F \wedge R) \vDash C_i$$

2. NEEDS_ASSUMPTION: The claim C_i is not strictly entailed by $(F \wedge R)$ alone, but requires an implicit premise A such that:
$$(F \wedge R \wedge A) \vDash C_i$$
Where A denotes a non-trivial, domain-grounded missing assumption.

3. UNSUPPORTED: The claim C_i directly contradicts $(F \wedge R)$ or represents an unbridgeable logical non sequitur.

When a claim is classified as NEEDS_ASSUMPTION, the system must extract the minimal assumption A, cite the target context gap (F_k or R_j), and normalize A into canonical legal predicate triple.

1.2 Scope Boundary: Logical Validity vs. Real World Truth

A critical formal boundary of this work is the distinction between validity and soundness. The system evaluates the logical consistency of an argument. It determines whether an argument claim is formally entailed by the stated context, if not, the system abduces a bridging premise A necessary to satisfy $(F \wedge R \wedge A) \vDash C$.

However, verifying the real-world empirical truthfulness of the abduced premise A is strictly out of scope. For example, in the classical syllogism where Socrates is a man and the conclusion is Socrates is mortal, abducing either "All men are mortal" or "All men are mortal and all women are immortal" creates a deductively valid syllogism. Our auto-formalization and verification framework confirms the logical necessity and formal validity of the link; it does not adjudicate external truth in the real world.

1.3 Research Questions

To evaluate this framework, the dissertation investigates three formal research questions:

- RQ1 (Pipeline Classification and Bias Mitigation): How effectively does a strict, ontology-grounded abductive pipeline mitigate the over-conservatism bias inherent in standard language models when verifying complex appellate reasoning?

- RQ2 (Deductive Failures and Artifact Detection): To what extent can high-sensitivity logical verification identify unsupported judicial leaps and expose structural data artifacts (such as rule circularity) within existing computational law benchmarks?
- RQ3 (Semantic Faithfulness & Ground-Truth Alignment): To what degree do auto-formalized canonical predicate triples preserve the substantive legal doctrines identified in human expert audits when evaluated via cross-encoder Natural Language Inference?

1.4 Dissertation Outline

- **Chapter 2 (Literature Survey)**: Reviews classical symbolic AI and law, statistical legal NLP benchmarks, the theoretical foundations of abduction, and auto-formalization architectures.
- **Chapter 3 (Methodology)**: Formulates the ontology-grounded abductive framework, details the canonical legal predicate vocabulary, and formalizes triple verbalization for neural verification.
- **Chapter 4 (Experiments & Evaluation)**: Quantifies pipeline precision and dataset artifacts across the full 250-case evaluation cohort, analyzes unsupported deductive failures, evaluates semantic fidelity via DeBERTa-v3 cross-encoder NLI, and audits structural flaws in the benchmark.
- **Chapter 5 (Conclusion and Future Work)**: Summarizes the principal research contributions, articulates project limitations, and outlines future directions for the "FIRAC-Abduce" benchmark.

Chapter 2: Survey

Legal adjudication has traditionally been formalized in computational jurisprudence as a rigid deductive derivation. Under this classical model, a court identifies an accepted set of material facts F, maps them onto established statutory rules or judicial precedents R, and derives a legally valid conclusion C, expressed formally as $(F \wedge R) \vdash C$. While early pioneering work in artificial intelligence and law successfully applied First-Order Logic (Kowalski and Sergot, 1986), Defeasible Logic (Prakken and Sartor, 1996), and Deontic Logic (von Wright, 1951; Sartor, 2005) to statutory texts, all these systems operated under the assumptions of a logically concluded knowledge base.

In practice, real-world judicial opinions rarely demonstrate mathematically complete deductive proofs. Instead, legal argumentation relies heavily on enthymemes – syllogistic deductions that rely on implicit, unstated premises assumed to be understood as common legal or factual knowledge. When classical deductive systems evaluate these incomplete arguments, they encounter logical discontinuities and incorrectly classify valid judicial decisions as fallacious or ungrounded leaps.

Enthymematic gaps in judicial discourse typically arise across three distinct dimensions:

- Factual Bridging: Connecting low-level evidence to statutory legal definitions (e.g., establishing that specific supervisory conditions make a worker an "employee" rather than an "independent contractor").
- Implicit Procedural & Constitutional Rules: Applying unstated procedural canons, constitutional standards of review (e.g., limits under Article 136 of the Indian Constitution, Res Judicata), or rules on statutory retroactivity.
- Commercial Common Sense: Relying on market customs, economic equivalence, and standard business practices that judges leave unwritten.

2.1 Modes of Logical Inference: Deduction, Induction and Abduction

Formalizing unstated assumptions requires distinguishing between the three classical modes of logical inference established by C.S. Peirce (Peirce, 1931):

1. Deduction: Derives necessary consequences from known premises. If the premises are true, the conclusion must be true:

$$A, A \rightarrow B \models B$$

2. Induction: Derives probabilistic generalizations from empirical observations:

$$\text{Observed } (A_1, B_1), (A_2, B_2), \ldots \Longrightarrow \forall x\big(A(x) \rightarrow B(x)\big)$$

3. Abduction: Inference to the best explanation (Kakas, Kowalski and Toni, 1992). Given an incomplete knowledge base K and an observed conclusion or judicial claim C, abduction infers the minimal, most plausible implicit premise A required to make the argument deductively valid:

$$\text{Given } K \text{ and } C, \text{ abduce } A \text{ such that } (K \wedge A) \vDash C$$

2.2 The Abduction-Space Problem & Legal Ontologies

When automated systems attempt abduction in natural language, they face the abduction-space problem (or vocabulary explosion). As natural language is open-ended, an unconstrained generative system can invent an infinite number of arbitrary, verbose explanations for any given gap.

To make abductive outputs computationally useful for formal reasoning and theorem proving, the hypothesis space must be constrained using a structured Legal Ontology. An ontology provides a formal, machine-readable vocabulary of legal entities, properties, and relations (Breuker et al., 2007; Sartor et al., 2011). In this work, extracted assumptions are constrained into canonical First-Order Logic (FOL) predicate triples:

$$[\text{Subject Concept}] \xrightarrow{\text{Legal Predicate}} [\text{Object Concept}]$$

This normalizes terms (e.g., standardizing concepts according to the Staram Indian Legal Ontology) and ensures semantic interoperability for subsequent symbolic verification.

2.3 Overview of Relevant Previous Work & Critical Evaluation

Existing approaches to automated legal reasoning, argument verification and premise extraction can be categorized into four primary paradigms:

1. Classical Symbolic AI & Law – First Order/Defeasible, mathematically sound, rigid and fails on gaps.
2. Argument Mining and Enthymeme Reconstruction
3. Statistical/LLMs – LegalBench, CaseHOLD, flexible with raw text, over-conservative bias.
4. Neuro-Symbolic Auto-Formalization – VeriCoT, LLM generates triples, formal ontology constraint.

2.3.1 Classical Symbolic AI & Formal Legal Logic

Early legal AI research focused on representing statutory rules and legal norms using formal logic frameworks, such as First-Order Logic, Defeasible Logic, and Deontic Logic (Kowalski and Sergot, 1986; Prakken and Sartor, 1996; Bench-Capon et al., 2012).

- Strengths: Symbolic engines provide deterministic proof trails, mathematical sound deduction, and complete auditability without hallucinations.

- Weaknesses: Hand-crafting formal rules from dense legal prose creates an intractable knowledge acquisition bottleneck. Crucially, symbolic systems operate under closed-world assumptions: when confronted with an enthymeme containing an unstated premise, the proof fails entirely.

2.3.2 Argument Mining and Enthymeme Reconstruction

Argument mining applies NLP to detect argument components (premises, claims) and relational structures (support, attack) from unstructured text (Mochales and Moens, 2011; Habernal and Gurevych, 2017). A specialized subfield focuses on enthymeme reconstruction – generating or retrieving missing premises in argumentative discourse (Rajendran, Bollegala and Parsons, 2016; Walton, 2001, 2008; Becker, Korfhage and Frank, 2021)

- Strengths: Explicitly models human argumentation as non-monotonic and enthymematic, analyzing broader multi-sentence discourse structures.
- Weaknesses: Existing work predominantly targets general domains (such as parliamentary debates or student essays) and lacks formal legal semantics. Additionally, reconstructed premises are generated as unconstrained natural language strings, making them unsuitable for formal symbolic verification.

2.3.3 Large Language Models & Legal Benchmarks

The emergence of Large Language Models (LLMs) like GPT-4, Llama-3, and domain-adapted transformers (e.g., Legal-BERT) has shifted the state-of-the-art across legal NLP tasks, including case summarization, statutory retrieval, and outcome prediction (Chalkidis et al., 2020; Katz et al., 2024; Guha et al., 2023).

Table 2.1: Critical Analysis of Existing Legal Verification Benchmarks:

Benchmark/ Dataset	**Primary Task Focus**	**Representation Structure**	**Critical Limitation for Abductive Legal Reasoning**
LegalBench (Guha et al., 2023)	Broad legal tasks across US law (contract review, statutory interpretation)	Multi-task classification, QA, and rule-matching	Evaluates isolated document-level tasks rather than multi-step judicial inference chains.
COLIEE (Rabelo et al., 2022)	Case law retrieval and legal entailment	Binary entailment (Y/N) on statute-case pairs	Treats entailment as a black-box binary classification; does not identify intermediate unstated premises.
CaseHOLD (Zheng et al., 2021)	Legal holding prediction from context	Multiple-choice Cloze QA	Tests language modeling of authoritative holdings without evaluating whether the intermediate deduction was complete.

FIRAC-Appeals	Sentence-level structural reasoning in Indian Supreme Court appeals	Segmented FIRAC blocks (Facts, Rules, Application steps)	Provides sequential application steps but lacks ground-truth labels for missing premises and contains circularity in extracted rule blocks.

Despite their strong contextual fluency, LLMs exhibit a fundamental vulnerability when auditing judicial reasoning: the over-conservatism bias (authority bias). Standard LLM verifiers routinely grant authoritative judicial phrasing the benefit of doubt, erroneously classifying incomplete enthymematic steps as deductively complete deductions without identifying the missing premises. Conversely, when prompted in an unconstrained manner to explain gaps, LLMs produce ungrounded hallucinations or default to cached template fallbacks.

2.3.4 Neuro-Symbolic Auto-Formalization & Verification (VeriCoT)

To combine the semantic fluency of LLMs with the logical rigor of symbolic systems, recent work has advanced Auto-Formalization and Verification Chain-of-Thought (VeriCoT) (Wu et al., 2022; Lyu et al., 2023; Feng et al., 2026). In these architectures, the LLM translates natural language text into a formal intermediate language (e.g., First-Order Logic, SMT-LIB, Prolog), which is then verified by a symbolic theorem prover (e.g., Z3 (De Moura and Bjørner, 2008), Prover9).

- Strengths: Decouples informal natural language parsing from formal proof checking, preventing logical hallucinations and ensuring mathematical transparency.
- Weaknesses: Existing auto-formalization pipelines are purely deductive; they evaluate whether a proof closes, but lack abductive mechanisms to hypothesize, categorize, and ground missing intermediate premises when an incomplete proof fails.

2.4 Research Gap and Summary

A comparative synthesis of the literature reveals a clear research gap:

1. Symbolic Systems provide verifiable deduction but fail when confronted with unstructured, enthymematic judicial text due to the knowledge acquisition bottleneck.
2. Standard LLM Verifiers parse natural language effectively but suffer from an over-conservatism bias that overlooks unstated assumptions, or an abduction-space explosion when unconstrained.
3. Current Legal Benchmarks evaluate document retrieval, binary entailment, or outcome prediction, offering no standardized framework for fine-grained, ontology-grounded legal abduction.

This dissertation addresses this gap by presenting a hybrid framework: a high-sensitivity LLM verification pipeline that detects logical discontinuities in judicial application steps, paired with a structured legal ontology (Staram Indian Legal Ontology) and deterministic Python guardrails to extract, classify, and formalize implicit legal premises into canonical First-Order Logic predicate triples.

Chapter 3: Methodology

This chapter builds on the formal task formulation in Section 1.1. It describes the architecture and the implementation of the ontology-based abductive verification system in detail. The methodology addresses two fundamental problems in computational legal analysis: the deductive over-conservatism of pre-trained language models on authoritative judicial prose, and the vocabulary explosion inherent in unconstrained natural language abduction.

3.1 Cumulative Entailment Architecture

The judicial reasoning in appellate court judgements does not just consist of disconnected memoryless deductions. While static verification models evaluate each judicial claim C_i purely against the initial evidentiary facts F and applicable statutory rules R, appellate court decisions construct sequential dependency chains. An intermediate judicial or factual conclusion deduced in the earlier step $k < i$ acts as a crucial legal premise for the step i.

To formalize this dynamic progression, we define the cumulative context state $\mathcal{K}_{i-1}$ at step i as:

$$\mathcal{K}_0 = F \cup R$$

$$\mathcal{K}_{i-1} = \mathcal{K}_0 \cup \bigcup_{j=1}^{i-1} \{C_j\}$$

Under memoryless evaluation ($F \wedge R \models C_i$), valid multi-step judicial claims are systematically misclassified as logical non sequiturs because intermediate holdings established along the inferential path are excluded from the working knowledge base.

In our architecture, the verifier evaluates formal entailment dynamically against the accumulated state $\mathcal{K}_{i-1}$. When the non-entailment condition $\mathcal{K}_{i-1} \not\models C_i$ is encountered, the abductive engine isolates the gap:

$$\mathcal{K}_{i-1} \wedge A_i \models C_i$$

where A_i is the smallest unstated proposition that needs to be preserved to sustain the chain of cumulative inference.

3.2 Pipeline Design and Deterministic Verification Mechanics

The verification framework decouples legal judgement auditing into three sequential stages: Context Serialization, High-Sensitivity Entailment Auditing, and Ontological Hypothesis Projection.

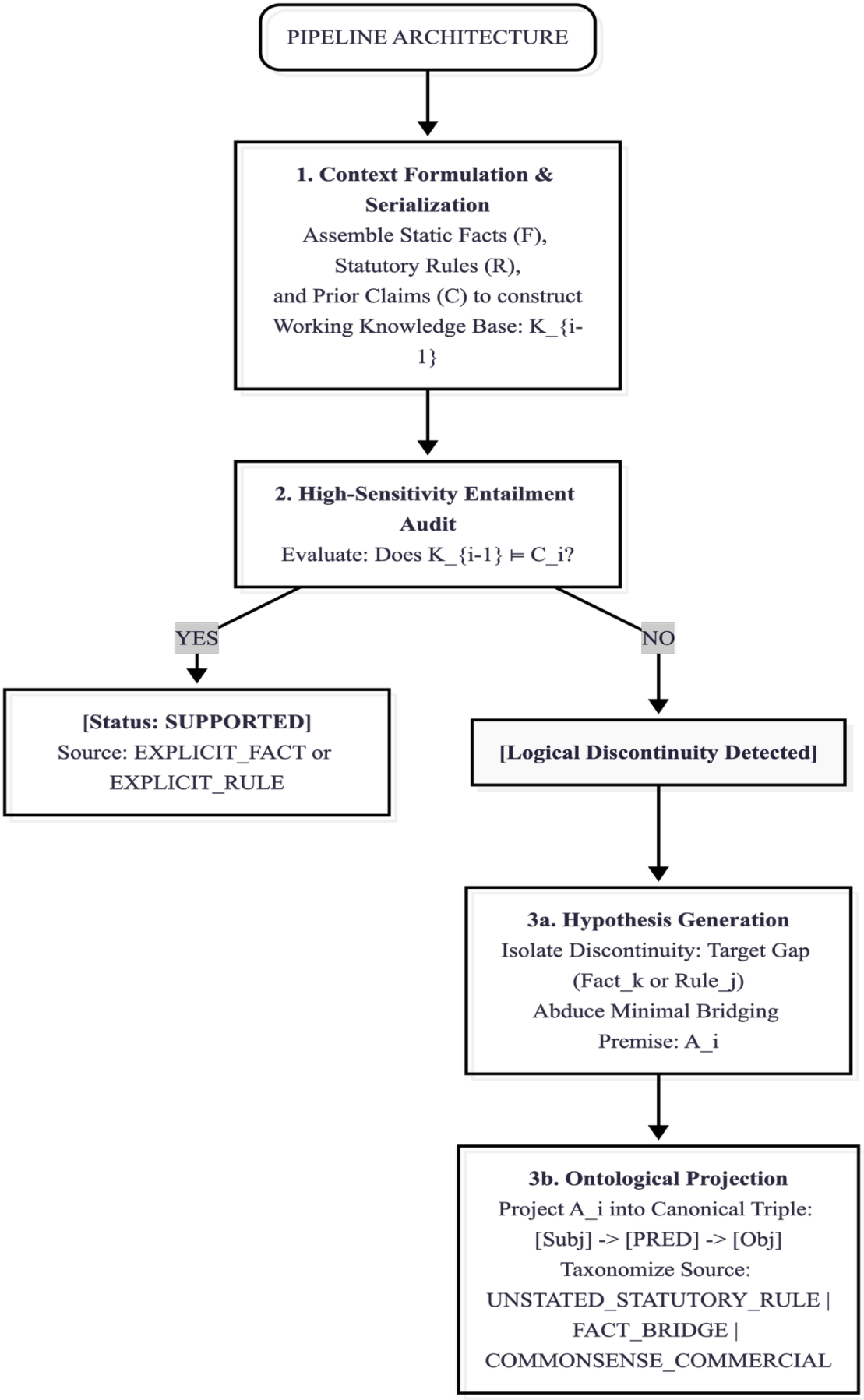


Figure 3.1: Overview of the cumulative entailment and abductive verification pipeline.

3.2.1 Stage 1: Context Serialization

The explicit evidentiary facts F and statutory rules R extracted from FIRAC representations are indexed and structured without conversational framing. For each step $i \in \{1, \ldots, k\}$, the prompt environment is dynamically constructed to include the cumulative reasoning history $\{C_1,...,C_{i-1}\}$ alongside static context.

3.2.2 Stage 2: High-Sensitivity Entailment Audit

Our initial prompting strategies were ineffective, with the models tending to give stronger evidentiary weight to prestigious judicial language. We mitigate this problem through our verification prompt where probabilistic and plausible reasoning are not allowed. The model must assess whether C_i is guaranteed with 100% formal certainty based exclusively on the explicit text in $\mathcal{K}_{i-1}$. If an inferential leap or unstated legal canon is required, the step must be classified as NEEDS_ASSUMPTION.

3.2.3 Stage 3: Ontological Hypothesis Projection

When detecting an inferential gap, the pipeline initiates abductive premise generation under a strict formal condition:

1. Target Gap Attribution: The system specifically cites the exact structural component ($f_k \in F$ or $r_j \in R$), where the inferential break occurs.
2. Canonical Triple Synthesis: The latent premise is projected into a First-Order Logic relational triple, bounded by a closed predicate schema.
3. Mutually Exclusive Source Taxonomy: The assumption is categorized into exactly one of three non-overlapping sources: UNSTATED_STATUTORY_RULE, COMMONSENSE_COMMERCIAL, or FACT_BRIDGE.

3.2.4 Deterministic Execution Controls

To eliminate generation stochasticity and ensure experimental reproducibility across runs:

1. Decoding Parameters: The verification engine enforces greedy decoding with temperature $\tau = 0.0$, top-$p = 1.0$, and fixed random seed initialization.
2. Deterministic Schema Validation: Outputs are validated against a formal JSON schema specification. The validator enforces syntactic compliance, confirms that the extracted relation belongs to the closed predicate set $P_{canonical}$, and rejects invalid or self-contradictory taxonomy states (such as classifying an assumption as originating from an explicit fact).

3.3 Propositional Step Segmentation

A major challenge in auditing legal text is sentence-level granularity. In original judicial judgements, a single grammatical sentence often contains multiple, distinct propositional inferences:

"The assessee, having failed to submit returns under Section 14 within the statutory window, cannot claim exemption under Section 4(3) because the factory was destroyed after the fiscal year ended."

This single sentence bundles three separate inferential operations:

1. A factual finding of non-submission of returns.
2. A procedural forfeiture of exemption rights.
3. An evidentiary finding regarding the timing of factory destruction.

To resolve this issue, the pipeline implements a Propositional Step Segmentation algorithm prior to verification:

- Rule-Based Clause Splitting: Sentences are split on subordinating and coordinating discourse markers ("because", "inasmuch as", "consequently", "whereas", "therefore").
- Propositional Atomicity Filter: Each clause is converted into an independent proposition $\tilde{C}_{i,j}$ containing exactly one main verb phrase and one legal assertion. This prevents the model from generating compound assumptions (e.g., merging procedural deadlines with substantive tax exemptions into an unformalizable sentence).

3.4 The Staram Legal Ontology and Canonical Predicate Schema

To resolve the abduction-space problem and prevent vocabulary explosion, extracted relational triples are strictly constrained to a finite, controlled set of formal relation types aligned with the Staram Indian Legal Ontology developed in this dissertation.

3.4.1 Empirical Derivation of the Vocabulary

To prevent test-set bias, this vocabulary was empirically derived using an independent development split of 25 training cases from the FIRAC-Appeals corpus. Unconstrained exploratory abduction generated a diverse set of raw relational phrases. Through qualitative thematic analysis, these raw phrases were manually reviewed, grouped by their core legal functions, and consolidated into 7 dominant jurisprudential categories.

3.4.2 Canonical Predicate Inventory

The resulting 7 canonical predicates are divided into two tiers:

- Meta-Level Relations: CONSTITUTES (definitional fulfillment), IS_EQUIVALENT_TO (economic equivalence), OVERRIDES (statutory precedence), REQUIRES_PREREQUISITE (jurisdictional trigger), EXCLUDES_APPLICATION (statutory invalidation).
- Operational Relations: INHERITS_NATURE_OF (factual substitution of asset character), ACTS_AS_SECURITY_FOR (commercial substance over form).

Table 3.1: Canonical Predicate inventory and formal logical templates.

Canonical Predicate	Tier	Formal Semantic Definition	Formal Logical Template
CONSTITUTES	Meta	Definitional satisfaction or legal status fulfillment.	$\forall x \left(\text{Subj}(x) \rightarrow \text{Obj}(x)\right)$

IS_EQUIVALENT_TO	Meta	Substantive commercial, economic, or legal equivalence between two distinct constructs.	$\forall x \left(\text{Subj}(x) \Leftrightarrow \text{Obj}(x)\right)$
OVERRIDES	Meta	Hierarchical statutory precedence (e.g., specific intent prevailing over general savings clauses).	$\forall x \left(\text{Subj}(x) \land \text{Obj}(x) \rightarrow \text{Apply}(\text{Subj})\right)$
REQUIRES_PREREQUISITE	Meta	Mandatory procedural condition or jurisdictional trigger necessary for legal validity.	$\forall x \left(\text{Valid}\left(\text{Subj}(x)\right) \rightarrow \text{Obj}(x)\right)$
EXCLUDES_APPLICATION	Meta	Statutory invalidation, forfeiture of privilege, or jurisdictional bar.	$\forall x \left(\text{Subj}(x) \rightarrow \neg\text{Permissible}\left(\text{Obj}(x)\right)\right)$
INHERITS_NATURE_OF	Operational	Factual substitution where compensation or recovery adopts the legal/tax character of the replaced item.	$\forall x, y \left(\text{Subj}(x) \land \text{Replaces}(x, y) \rightarrow \text{Type}\left(x,\text{Type}(y)\right)\right)$
ACTS_AS_SECURITY_FOR	Operational	Substantive commercial intent overriding formal contract nomenclature (substance over form).	$\forall x \left(\text{Subj}(x) \rightarrow \text{CommercialFunction}(x,\text{Obj})\right)$

3.5 Deterministic Triple Verbalization

As raw bracketed triples ([Subject] → [Predicate] → [Object]) represent an out-of-distribution artifact for neural Natural Language Inference (NLI) models, we implement a Deterministic Verbalization Operator $\mathcal{V}(\tau)$. This operator applies a fixed grammatical copula to the formal predicate (e.g., mapping OVERRIDES to the string "overrides"), converting the symbolic triple into a fluent, machine-verifiable English assertion for downstream semantic evaluation.

Chapter 4: Experiments and Empirical Evaluation

To evaluate the performance of the abductive legal reasoning pipeline, the empirical evaluation investigates three formal research questions (RQs) using a two-tier evaluation strategy. Tier 1 executes an automated quantitative evaluation across the 250-case evaluation cohort to analyze structural verification dynamics and schema compliance. Tier 2 conducts a fine-grained semantic Natural Language Inference (NLI) audit on a gold-standard manual cohort to verify the qualitative legal fidelity of the extracted premises.

4.1 Experimental Corpus and Compute Environment

The automated verification was conducted across a stratified cohort of 250 appellate decisions from the FIRAC-Appeals benchmark. The quantitative evaluation encompassed all 3,534 reasoning steps within this evaluation cohort and was performed using qwen-2.5-coder-7b-instruct.

4.2 RQ1: Pipeline Classification and Dataset Artifacts

Across the full corpus of 3,534 reasoning steps, the pipeline yielded the following classification distribution:

- 3,168 steps (89.64%) were classified as SUPPORTED.
- 252 steps (7.13%) were classified as UNSUPPORTED (logical leaps).
- 114 steps (3.23%) were classified as NEEDS_ASSUMPTION, split between UNSTATED_STATUTORY_RULE (72), COMMONSENSE_COMMERCIAL (41), and FACT_BRIDGE (1) sources.

Of the 114 extracted premises, 105 (92.1%) used one of the seven canonical predicates on the first valid attempt or after retry, with the following distribution: CONSTITUTES (42), IS_EQUIVALENT_TO (31), INHERITS_NATURE_OF (17), EXCLUDES_APPLICATION (11), OVERRIDES (2), REQUIRES_PREREQUISITE (2), and ACTS_AS_SECURITY_FOR (0). The remaining 9 cases (7.9%) exhausted the retry budget without producing a schema-compliant triple; these are discussed as a limitation in section 5.2.

4.3 RQ2: Deductive Failures and Ungrounded Leap Detection

While structural artifacts in the dataset (such as rule circularity, discussed in detail in Section 4.5) suppressed the extraction of implicit assumptions, the pipeline successfully isolated deductive failures. Across the full corpus, the model flagged 252 claims as UNSUPPORTED – instances where the court's stated reasoning lacked explicit factual or statutory backing, creating a logical discontinuity rather than a direct textual contradiction.

- Procedural Overreach: In labor dispute in_abs_train_1007, the model rejected the court's inferential leap regarding interim relief, explicitly stating in its explanation that "interim relief does not guarantee final relief".
- Hallucinated Precedents: In case in_abs_train_1065, the pipeline flagged four consecutive judicial claims as UNSUPPORTED. It accurately noted that there was "no explicit statement in the facts or rules about the Supreme Court reviewing the test applied in Raghuvanshi Mills".
- Evidentiary Gaps: In case in_abs_train_104, the model flagged a sequence of four claims as UNSUPPORTED because the narrative introduced new elements regarding "disagreement" and "hypothetical actions" that were entirely absent from the established dispute record.

In case in_abs_train_1175, the system identified a missing procedural-fairness gap, extracting the canonical triple [Custodian General] -> [REQUIRES_PREREQUISITE] -> [reasonable opportunity of being heard] and correctly classifying its source as COMMONSENSE_COMMERCIAL.

4.4 RQ3: Semantic Alignment Verification via Cross-Encoder NLI (DeBERTa-v3)

To evaluate the qualitative legal accuracy of extracted premises beyond categorical agreement metrics, a gold-standard cohort of 5 cases was manually reviewed by the author and a second legal reviewer. For each case, the reviewers established a human ground-truth premise (P) from the original judgement text, which was then compared against the pipeline's verbalized canonical triple (H) using a pretrained cross-encoder natural language inference model, cross-encoder/nli-deberta-v3-base (He, Gao and Chen, 2021).

Table 4.1: Cross-encoder NLI evaluation of manually audited cases.

Case ID	Legal Domain	Human Ground Truth Premise (P)	Verbalized Triple (H)	NLI Prediction	P (Entailment)
in_abs_train_1028	Tax Notice	Underlying tax not officially assessed prior to notice.	Serving a notice under Section 46 requires a prior official tax assessment as a prerequisite.	Entailment	0.871
in_abs_train_1041	Goodwill	Hospitality standards dictate goodwill beyond physical lease.	Theatre goodwill inherits the nature of customer reputation rather than leasehold value.	Contradiction *	0.448

in_abs_train_1016	Title Retention	Installment title retention acts as sale security, not hire.	Retaining machinery title during an installment contract acts as security for a deferred sale.	Entailment	0.990
in_abs_train_103	Indemnity	Indemnity replacing lost gross profits inherits taxable nature.	An insurance indemnity received for lost profits inherits the nature of taxable business profit.	Entailment	0.993
in_abs_train_1023	Interim Relief	Final relief at interim stage is fatal legal error.	Awarding complete job reinstatement as interim relief excludes the application of interlocutory relief.	Entailment	0.999

*Lexical Sensitivity: in_abs_train_1041 was flagged as Contradiction due to cross-encoder sensitivity to the comparative phrasing ("rather than" vs. "beyond"), despite consistent jurisprudential intent.

Across all 5 cases, the average entailment probability is 86.02%, with four of five cases yielding strong entailment ($P > 0.87$) and one flagged contradiction attributable to surface-level lexical sensitivity rather than substantive legal disagreement.

4.4.1 Qualitative Findings from the Manual Audit

Across the five manually audited cases, two distinct categories of enthymematic gap emerged, formally distinguishable using the notation from section 1.1. In three cases (1028, 1041, 1023), the gap was one of incompleteness – the stated Facts and Rules were insufficient to derive the Conclusion ($\text{Facts} \wedge \text{Rules} \nvdash \text{Conclusion}$) requiring an additional premise to bridge a genuine informational absence (e.g., case 1028's Factual Assessment Status Gap, where the record simply never states whether an assessment occurred). In two cases (1016, 103), by contrast, the literal text of the Facts and Rules appeared to contradict the court's Conclusion ($\text{Facts} \wedge \text{Rules} \vdash \neg\text{Conclusion}$), and the correct assumption instead reinterpreted a fact's legal character to resolve the apparent contradiction (e.g., case 1016's Structural Installment Contract Gap, where title retention textually resembles a non-sale but functions as a financing security once "sale" is understood substantively rather than formally). This distinction – incompleteness gaps versus substance-over-form reinterpretation gaps – suggests the pipeline's single NEEDS_ASSUMPTION category may be masking two logically distinct abductive tasks, worth separating in future annotation schemes.

4.5 Systematic Audit of Dataset Flaws in the Benchmark

Auditing the FIRAC-Appeals benchmark against the automated pipeline output and original court records revealed systemic structural anomalies that artificially affect model performance:

- Rule Circularity: In a significant portion of cases, the Rules section contains a summary of the court's final holding rather than pre-existing statutes. Because the conclusion is pasted directly into the input premises, the verifier marks the reasoning step as directly SUPPORTED, masking genuine enthymematic leaps.
- Fact Summarization Artifacts: Aggressive dataset summary missed certain evidentiary points at different locations, making seemingly nonexistent "judicial leaps" which were then flagged as UNSUPPORTED by the pipeline.
- Procedural Omissions: Foundational principles (e.g., limits of Article 136, res judicata) were routinely omitted from explicit rule annotations.
- Compound Propositional Granularity: Each chain of reasoning was broken down at grammatical sentence limits rather than atomic logical clauses, occasionally prompting compound evaluations to be executed in a single step.

Chapter 5: Conclusion

5.1 Project Summary

Automated legal reasoning systems cannot depend on clear step-by-step deductive reasoning. In real-life judicial decisions, judges often present arguments in the form of enthymemes – arguments that omit intermediate factual steps, commercial practices, and background statutory doctrines as they are assumed to be well-known legal rules. Although Large Language Models can understand natural language well, baseline verifiers suffer from an over-conservatism bias, routinely accepting incomplete judicial reasoning chains as deductively complete.

To address this challenge, this dissertation developed and evaluated a high-sensitivity, ontology-grounded abductive verification framework tested across 250 Indian Supreme Court appeal cases (3,534 reasoning steps) from the FIRAC-Appeals dataset. By formulating project findings around the core research questions, the primary contributions and answers established by this work include:

1. Answering RQ1 (Pipeline Classification & Bias Mitigation): The proposed pipeline successfully mitigates model over-conservatism by enforcing strict formal entailment and ontological constraints. Evaluated across the full benchmark cohort, it achieved an 89.64% SUPPORTED classification rate, isolated 252 unsupported logical leaps (7.13%), and successfully extracted and grounded 114 implicit legal premises (achieving 92.1% canonical-predicate compliance).

2. Answering RQ2 (Deductive Failures & Artifact Detection): High-sensitivity logical verification proved highly effective at exposing structural data artifacts within computational law corpora. Beyond isolating unsupported judicial leaps, the pipeline systematically uncovered critical benchmark flaws—most notably rule circularity and fact summarization artifacts—providing empirical justification for future re-annotated benchmark guidelines ("FIRAC-Abduce").

3. Answering RQ3 (Semantic Faithfulness & Ground-Truth Alignment): Cross-encoder Natural Language Inference (NLI) validation against a manual gold-standard audit confirmed that auto-formalized canonical predicate triples preserve substantive legal doctrines with high fidelity, achieving an average entailment probability of 86.02% against expert human ground truth.

4. Deterministic Pipeline Robustness & Ontological Grounding: Methodologically, the framework eliminated taxonomy contradictions entirely (0.0% clash rate) and solved the unconstrained abduction-space problem by bounding all extracted assumptions into closed canonical predicate triples ([Subject] -> [Predicate] -> [Object]) aligned with the Staram Indian Legal Ontology.

5.2 Limitations

Despite these achievements, several limitations remain:

- Dataset Artifact Interference: Structural flaws in the FIRAC-Appeals dataset corpus – particularly the inclusion of final court holdings inside explicit rule blocks – occasionally presented as genuine enthymemes through circular reasoning.

- Single Jurisdiction Scope: The evaluation and ontology schema were grounded specifically in Indian Supreme Court jurisprudence (statutory tax, corporate, and constitutional appeals), and their generalizability across common law or civil law jurisdictions remains to be tested.

- Sentence-Level Granularity: The dataset parts judicial reasoning along grammatical sentence boundaries rather than atomic propositional clauses, occasionally requiring the pipeline to extract compound assumptions for a single step.

- Verification of Real-World Truth vs. Formal Entailment: The pipeline is designed to test formal internal consistency rather than external ground truth. When the system extracts a missing premise A such that $(F \wedge R \wedge A) \vDash C$, it guarantees that A is sufficient to bridge the deductive gap within the closed reasoning chain. It does not – and cannot – verify whether A is factually or empirically true in the real world. Validating the real-world factual accuracy of abduced premises remains an accepted boundary of this study and a designated direction for future work.

- Predicate Compliance Ceiling: During the 250-case automated run, 9 of the 114 extracted premises (7.9%) failed to produce a schema-compliant canonical predicate after exhausting the retry budget, instead defaulting to either a non-canonical free-text predicate (6 cases) or an unstructured restatement of the claim (3 cases). This reflects a residual limitation of enforcing a closed ontological vocabulary on a 7-billion-parameter model under a fixed retry ceiling, rather than a flaw in the ontology itself.

Appendix A: Verification Prompts and Canonical Predicate Schema

A.1 System Verification Prompt

The following system prompt enforces a strict output structure, a classification system that does not overlap and a canonical predicate triple structure without conversational preambles:

You are a strict formal legal verifier evaluating judicial reasoning steps (Application claims) against explicitly stated Facts and Rules to extract UNSTATED IMPLICIT PREMISES (abductive logic).

CRITICAL FORMAT & COMPLETENESS RULES:
1. COMPLETENESS (NO TRUNCATION): You MUST process EVERY SINGLE claim listed in the user prompt sequentially from Step 1 to the final step. Do NOT stop early, do NOT summarize, and NEVER output notes like "I stopped at Step X".
2. NO PREAMBLE: Output ONLY the raw valid JSON array starting with '[' and ending with ']'. No introductory or conversational phrases.
3. CASE-SPECIFIC PREMISES: The extracted triple MUST use entities and concepts explicitly present in the CURRENT case context. NEVER fallback to generic example terms (such as 'JudgementDebt', 'Salary', or 'Decree') unless explicitly in the case.

TAXONOMY CLASSIFICATION:
- SUPPORTED: Follows with 100% strict formal deduction from stated Facts or Rules alone.
- NEEDS_ASSUMPTION: Relies on an unstated premise to bridge a gap between Facts, Rules, or Conclusions.
- UNSUPPORTED: Contradicts stated Facts/Rules or is an illogical leap.

SOURCE OF PREMISE (`source_of_premise`):
- If SUPPORTED: Use 'EXPLICIT_FACT' or 'EXPLICIT_RULE'.
- If NEEDS_ASSUMPTION: Must use one of: 'UNSTATED_STATUTORY_RULE', 'COMMONSENSE_COMMERCIAL', or 'FACT_BRIDGE'.

ONTOLOGY TRIPLE SCHEMA (`missing_premise`):
Format extracted assumptions strictly as: "[Subject Concept] -> [Legal Predicate] -> [Object Concept]"

JSON OUTPUT SCHEMA:

```
[
 {
   "step_index": 1,
   "claim": "<exact claim text>",
   "status": "<SUPPORTED | NEEDS_ASSUMPTION | UNSUPPORTED>",
   "target_gap": "<cite exact Fact_X or Rule_Y ID where gap occurs, or 'none'>",
   "missing_premise": "<[Subject] -> [Legal Predicate] -> [Object], or 'none'>",
   "source_of_premise": "<EXPLICIT_FACT | EXPLICIT_RULE | UNSTATED_STATUTORY_RULE | COMMONSENSE_COMMERCIAL |
```

FACT_BRIDGE>",
  "explanation": "<one precise sentence explaining the formal logical link or gap>"
}]

A.2 User Prompt Template

EXPLICIT FACTS:
{facts_joined}

EXPLICIT RULES:
{rules_joined}

REASONING CLAIMS TO CHECK (YOU MUST PROCESS ALL {total_claims} STEPS):
{claims_formatted}

Analyze ALL {total_claims} claims now. Return ONLY the valid JSON array.

Appendix B: Manual Inspection Logs for Benchmark Cases

Case_ID	Contents	Abductive Analysis & Notes
in_abs_train_1028 – Tax Notice	• Facts: ITO served a notice under Sec 46(5) to deduct tax from a civil decree. • Rules: Sec 46(5) requires tax to be officially assessed and remain unpaid. • Conclusion: Notice invalid; appeal dismissed.	• Initial State: Facts ∧ Rules ⊬ Conclusion (The rule lacks its factual triggering condition). • Gap: Factual Assessment Status Gap (Explicit facts omit whether assessment occurred). • Assumption (A): "The underlying income tax had not been officially assessed."
in_abs_train_1041 – Goodwill	• Facts: Tribunal valued theatre goodwill at Rs. 2 lakhs based strictly on its physical lease. • Rules: Goodwill must incorporate broader reputation and connection (*Trego*). • Conclusion: Erred in law; appeal allowed.	• Initial State: Facts ∧ Rules ⊬ Conclusion (A computed valuation is textually still a value). • Gap: Operational Business Nature Gap (Text ignores intangible service industry traits). • Assumption (A): "Hospitality/service standards dictate theatre goodwill, independent of lease value."

in_abs_train_1016 – Dam Contract	• Facts: Corporation supplied machinery to contractors; explicitly retained legal title until full payment. • Rules: Sale requires ownership transfer; hire permits return at will (Helby). • Conclusion: Transaction is a sale; appeal dismissed.	• Initial State: Facts ∧ Rule ⊢ ¬Conclusion (Contradiction: Title retention textually equals no sale). • Gap: Structural Installment Contract Gap (Title retention functions as a financing security). • Assumption (*A*): "Non-cancellable installment title retention acts as a deferred sale security, not a hire."
in_abs_train_103 – Mill Fire	• Facts: Mill received a Rs. 14,00,000 lump-sum insurance payouts after a catastrophic fire. • Rules: Sec 4(3)(vii) exempts casual, non-recurring receipts from tax. • Conclusion: Payout is taxable income; appeal dismissed.	• Initial State: Facts ∧ Rules ⊢ ¬Conclusion (Contradiction: A fire accident is textually casual/rare). • Gap: Accounting Substitution Gap (Payout explicitly replaces standard taxable revenue). • Assumption (A): "An indemnity calculated to replace lost gross profit inherits the taxable nature of business profit."
in_abs_train_1023 – Labor Shift	• Facts: Worker fired without permission during a pending dispute around an amendment timeline. • Rules: Tribunal granted full job reinstatement as a temporary interim relief measure. • Conclusion: Interim order set aside; appeal allowed.	• Initial State: Facts ∧ Rules ⊬ Conclusion (Rule mismatch leaves core timeline dispute unresolved). • Gap: Remedial Equivalence Gap (Fatal lower court procedural error overrides substantive debate). • Assumption (A): "Awarding final relief at an interim stage is a fatal legal error bypassing background statutory dispute."

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